# MANOSREE N

Software Engineer · AI-Enabled Product Development
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## SUMMARY

Third-year IT undergraduate with strong Data Structures, OOP and full-stack fundamentals, building AI-powered products end to end, from problem definition to deployment. Shipped a Claude-powered enterprise data-pipeline UI for a manufacturing client, built a multilingual RAG system for Smart India Hackathon, and published peer-reviewed research on explainable AI. Uses AI coding tools to move fast while validating their output with sound engineering judgment. Seeking a Software Development Engineer internship.

## CORE SKILLS

**CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Systems
**Languages:** Python, Java, C, C++, JavaScript, SQL
**Web & Backend:** FastAPI, Flask, React, Next.js, REST APIs, WebSockets, HTML/CSS/JavaScript
**Databases & Tools:** PostgreSQL, SQLite, DuckDB, Linux, Nginx, Git, Docker, AWS, CI/CD, pytest, Postman
**AI / ML:** LLM APIs, RAG Pipelines, LangChain, Prompt Engineering, PyTorch, TensorFlow, Scikit-learn, Hugging Face, SHAP

## EXPERIENCE

**Admin Console Developer — Woven Intelligence, SRM Textiles** *Aug 2026 – Present*
*React, Next.js, WebSockets, PostgreSQL, DuckDB*

- Owned the Schema Governance UI and Human-in-the-Loop review queue for a Claude-powered enterprise data pipeline, enabling real-time validation and single-click anomaly resolution.
- Built real-time processing-status updates over WebSockets and role-based access control for separate plant-manager and operator views.

**Full-Stack Development Intern — Technostack, India** *Jul 2025 – Aug 2025*

- Reduced backend query response times by 40% by building scalable, database-optimized architectures in Python, Flask and RESTful APIs.
- Cut deployment delivery time by 30% by implementing automated CI/CD pipelines for testing and release.
- Integrated ML models into production services, taking features from prototype to deployed, end-to-end endpoints.
- Delivered responsive, accessible UIs in HTML, CSS and JavaScript, collaborating with design and product teams.

## PROJECTS

**Real-Time Server Health Dashboard — Linux Telemetry & Alerting System**
*Python, Flask, Gunicorn, SQLite (WAL), psutil, Chart.js, Nginx, systemd, pytest | github.com/Manosree30/Server_monitor*

- Built a monitoring service where a background collector thread samples CPU, memory and disk into SQLite (WAL mode), so API endpoints only serve pre-sampled data and never block on system calls.
- Designed an alert state machine with debounce (3 consecutive breaches) and hysteresis (5-point recovery margin) to eliminate false alarms and alert flapping, with full alert lifecycle tracking.
- Enforced a single-collector design (one Gunicorn worker plus file lock) to avoid duplicate samples and race conditions; added data-retention pruning, unit tests, a stress-testing tool capped at 80% RAM, and a systemd + Nginx basic-auth deployment.

**AyurGuard — Multilingual RAG Assistant for Ayurveda IP & Regulatory Guidance** *Sep 2026*
*React/Next.js, FastAPI, Vector DB, Hybrid Search (Vector + BM25) | Smart India Hackathon (SIH26045)*

- Identified that practitioners lack reliable, cited answers across scattered sources, and built a hybrid RAG system (vector + BM25 keyword search) that retrieves and cites AYUSH, WIPO, TKDL and WHO documents.
- Designed citation-verification logic that checks answers against retrieved sources to reduce hallucination, with multilingual query understanding.

## RESEARCH PUBLICATIONS

- “A SHAP-Driven Explainable AI Approach for Predicting and Interpreting River Pollution Risk Using Ensemble Learning.” Manosree N. Proceedings of the 18th International Conference on Science and Innovative Engineering (ICSIE 2026), October 2026.

## EDUCATION

**B.Tech, Information Technology — Bannari Amman Institute of Technology, Tamil Nadu** *Expected 05/2028*
*GPA: 8.42 | Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Machine Learning, Deep Learning*

## CERTIFICATIONS

- Full Stack Development — Technohacks (2025); Cloud Infrastructure Foundations — Oracle University (2024)
- Data Science Contest — XPECTO’26, IIT Mandi

## ACHIEVEMENTS & RECOGNITION

- Finalist: Guidewire Hackathon Phase 2, Synaptix Frontier AI Hack, Inter-College Coding Hackathon (2026)
- Winner, BIT Poster Presentation (2025)
- Shortlisted (Smart Waste AI), Cyfuture AI Hackathon — Govt. of Madhya Pradesh, Ujjain Mahakumbh (2024)